# Final Project

In your final project, you and your team members will implement an information system (or software, or program, whatever you prefer; in this document, we will call it a system).

Your system may contain one or more programs, modules and functions. In addition to being able to use **Python**. There are basically no restrictions except completion[1], which means that other programming languages cannot be used. As for your programs, modules and functions, you can develop them by yourself, use third-party ones directly, obtain them through Vibe Coding, or modify them provided by AI.

The final topic is what you can expect
(1) Apply the project topic in an interesting yet challenging scenario;
(2) Analyze the problem and develop a good structure and division of labor[2];
(3) Write correct, effective, and maintainable Python programs;
(4) Try to write some of the findings from the above process into a project report, and
(5) Present your ideas and outcomes clearly and engagingly in up to 12 minutes in the 16th Week[3].

In addition, because this is the final project, some details are left to you to decide. During the design and development process, you are welcome to discuss with everyone at NTU COOL, or talk to the teaching assistants or Tommy. You can

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[1] It is relatively painful to create a beautiful UI interface with Python (it involves art design), so you don't need to spend lots of time on art design for the final project.
[2] The division of labor can be based on modules or functions, but it does not include recording videos, making presentations, and writing project reports, as these three tasks should be done by the entire team together.
[3] If needed, you may record a video as supplementary material for the whole class to refer to.

find reference materials on your own during the idea generation process, but **CANNOT directly use** their solutions or only change a little bit. If you are found out in this situation, you will get **Zero points**. You will likely need a lot of knowledge and skills that weren't taught in class. If this happens, please use whatever method you have to teach yourself; believe in yourself, you can do it!

I hope you all enjoy working on your final project. Good luck!

# Project Theme:
# Design an explainable AI decision support system that helps users make judgments under real-world data constraints, rather than simply providing a single prediction value.

Plain Explanation:

What you need to build is a small but complete machine learning prediction system (AI). The system must:

1. Have a clear use scenario
2. Use AI to assist humans in making decisions
   (Who are you helping? What decision are they making?)
3. Explain why the AI makes its judgment
4. Clearly explain why you designed the system this way rather than using other approaches
   (How do you help people understand AI?)

Examples (Not Limited to These — Please Be Creative):

1. Explainable Housing Rent Evaluation Decision Support System

    I. What can the system do?

    - Allow users to input: location, floor area, building age, floor level, other relevant factors
    - The system outputs:
        - Recommended rent range
        - Top three factors influencing the rent
        - Explanation if the rent is considered too high
            - Which condition is pushing the rent higher?

    II. AI should not only output a number

    - Instead of only saying: “Recommended rent: 28,000”

- The system should explain:
    - because location and floor area increase the price,
    - but building age lowers the price,
    - therefore the rent falls within this range.

2. Explainable AI Decision Support System for Course Enrollment and Grade Risk

I. What can the system do?

- The system allows users to input or retrieve:
    - Completed assignments
    - Quizzes
    - Attendance (if available)
    - Midterm score (if available)
    - Course grading policy (weight distribution)
- System Outputs (Three Types of Information):
    - Current Status Assessment
        - If the student drops the course now, what will happen?
        - If the student continues the course, under different effort scenarios:
            - Possible grade ranges (not guaranteed outcomes)
    - Risk Explanation
        - What is the largest source of risk?
            - Assignments?
            - Attendance?
            - Exams?
        - Which factors have the greatest impact on the final grade?
    - Action Recommendations (Not Guarantees)
        - If the goal is to pass the course
        - If the goal is to achieve B+ or higher
        - The system should explain:
            - Which actions have the greatest impact
            - Which efforts have low marginal benefit

II. AI should not only output a number

- Instead of only saying: “Final grade: 72”
- The system should explain:
    - because the current assignment average is 85, which raises the overall grade,
    - but the midterm score is only 62, which lowers the grade,
    - therefore the system estimates:
        - the final grade may fall between 68~78 (a probabilistic range estimate, not a guarantee).
- Current Status Assessment
    - If the student drops the course now:
        - The student will not receive credit this semester and must retake the course next semester.
    - If the student continues the course, under different effort scenarios:
        - Maintain current effort level (assignments around 80~85, normal attendance)
            - Possible final grade: 68~72
        - Improve assignment quality and maintain attendance (assignments above 90 for the remaining ones)
            - Possible final grade: 72~80
        - Improve overall performance (assignments above 90 and final exam above 80)
            - Possible final grade: 78~85
        - These are probabilistic ranges, not guaranteed outcomes.
- Risk Explanation
    - The main sources of risk are:
        - Low exam performance (midterm score: 62)
        - Some late assignments
        - Attendance rate around 70%
    - System explanation:

        - because exams account for 40% of the total grade and the current score is relatively low,
        - but assignments still have room to improve to above 90,
        - therefore final exam performance is the most critical factor affecting the final grade.
- Action Recommendations (Not Guarantees)
    - If the goal is to pass the course (60 points)
        - Ensure all remaining assignments are submitted on time
        - Increase attendance to above 80%
        - Achieve at least 65 on the final exam
    - If the goal is to achieve B+ or higher (85 points)
        - Maintain assignment averages above 90
        - Maintain attendance above 90%
        - Achieve at least 85 on the final exam
- System Explanation
    - The system analysis indicates:
        - Actions with the greatest impact
            - Improving final exam score
            - Ensuring all assignments are submitted on time
        - Efforts with lower marginal benefit
            - Increasing assignment scores from 90 to 95 (limited impact on the final grade)
            - therefore, the system recommends prioritizing: final exam preparation and timely assignment submission.

Important Note

All final project systems created in this course are considered Decision Support Systems. The AI component only provides analysis, risk evaluation, and action recommendations. It does not constitute a prediction guarantee, final decision, or behavioral commitment.

# Detailed Project Guidelines

All groups must follow the requirements below when developing the system, conducting research, writing the report, and producing videos (if needed).

All materials (project report, slides, code, etc.) must comply with these guidelines.

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### 1. Team Formation and Division of Work

- **Team Size:** each group must consist of 3~5 students.

- **Division of Work:** Each group should clearly assign responsibilities, including: data collection, data preprocessing, model development, result analysis, report writing, presentation preparation, etc. The presentation must specify each member's contribution.

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### 2. Research Topic and Content Requirements

- **Main Theme:** Explainable AI Decision Support Systems
- **Topic:** No restriction
- **Scope:** No restriction
- **Practicality, Applicability, and Business Value:** The report should discuss: real-world application scenarios, potential business value, and possible future extensions of the AI system

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### 3. Report Format and Documentation Requirements

- **Required Template:** The final project report must follow the "Python DM Report Template." All groups must download and use this template to ensure a consistent report format.

- **Format Requirements:**

- Language: English or Chinese (choose one)
- Sections may be slightly adjusted, but the overall structure must follow the template.
- Reports must include:
  - Figures
  - Flowcharts
  - Experimental results
- All figures must have titles and descriptions.

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## 4. Important Dates and Tasks

| Week | Task |
|---|---|
| Week 5 | Final project announcement<br>Upload the initial discussion draft (any format) to the assignment system |
| Before Week 16 class | Each group uploads all project materials to NTU Cool[4]<br>Materials include (but are not limited to):<br>• Slides<br>• All project code (must be executable) and README<br>• Final report (PDF + Word or LaTeX)<br>• All editable files (e.g., figures)<br>• Video explanation (optional) |
| Within two days after the Week 16 class, | Every student must submit:<br>• Evaluation and feedback for other groups<br>• Internal team peer evaluation<br><br>Deadlines: |

[4] If the storage space on NTU Cool is not sufficient to upload all the materials, please first upload the files to your own cloud storage and then share the link on NTU Cool. Make sure your teammates test the link to confirm that all files can be accessed and downloaded completely without issues. Uploading the project to GitHub is also recommended.

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|---|---|
| before 11:59 PM | ➔Classes 01 & 02: June 10, 11:59 PM<br>➔Classes 03 & 04: June 11, 11:59 PM |

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**5. Additional Considerations**

- **Data Ethics & Privacy**: When using social media data, ensure strict adherence to the platforms' data usage policies and protect user privacy.
- **Citation & Copyright**: All cited references, figures, and external resources must have proper attribution and adhere to academic integrity.
- **Queries & Support**: If you encounter technical or research challenges during the project, please reach out to teaching assistants or Tommy promptly to ensure the project remains on track.

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**6. Academic Paper vs Technical Report**

Both academic papers and technical reports are forms of professional documentation.

Similarities:

- Purpose
  Both communicate specific knowledge, findings, or technical results.
- Structure
  Both usually contain structured sections such as:
  - Abstract
  - Introduction
  - Problem statement
  - Methodology
  - Results
  - Discussion

Differences:

- Target Audience
  Academic papers mainly target the research community.
  Technical reports focus more on practical applications in professional fields.
- Writing Style
  Academic papers emphasize formality and academic rigor.
  Technical reports emphasize clear technical descriptions and practical application.
- References
  Academic papers usually contain extensive citations and literature review.
  Technical reports may emphasize practical cases and experimental results.

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We encourage creativity, rigorous analysis, and a strong fusion of theory and practical application.
If you have any questions regarding these guidelines or require further clarification, please contact Tommy as soon as possible.
Good luck, and we look forward to seeing your innovative projects achieve high quality!